



Dynamic simulation of natural gas pipeline network based on interpretable machine learning model

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ABSTRACT

Natural gas pipeline network modeling and simulation is the basis of dispatch and design. Modeling methods based on the mechanistic model have for a long time been facing the problem of multi-parameters and multi-flow patterns that are difficult to determine. Additionally, the method of purely machine learning has the problems of poor interpretability and difficulty in optimizing the model. A novel dynamic simulation method based on an interpretable shortcut Elman network (Shortcut-ENN) model for the pipeline network is proposed. The Shortcut-ENN model is derived from the state space equations. Based on the Shortcut-ENN model, the connection relationship and mechanism characteristics of the pipeline are retained, and an interpretable machine learning pipeline network model is constructed to make up for the lack of mechanism modeling. The model fully adopts the mechanism knowledge and is very suitable for optimization, which greatly improves robustness of the model. Validated and compared with long short-term memory model, the results show that MSE , MAE , R^2 , and EV of the proposed Shortcut-ENN-based model considering embedded pipeline mechanism and compressor constraints are improved approximately 84.4%, 60.1%, 0.75%, and 53.3%, respectively, and the R^2 is about larger than 0.99, and the EV is about less than 0.02.

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1. Introduction

Natural gas is an important clean energy in the world and an important resource that has a profound effect on the sustainable development of society [1,2]. As of the end of 2019, natural gas consumption had reached 3.93 trillion cubic meters, and its share in primary energy had risen to a record high of 24.2% [3]. Long-distance pipelines are important carriers for natural gas transportation [2,4]. The total length of the current trunk/transmission pipeline network is 2,034,065.0 km, of which the natural gas pipeline constitutes 1,295,563.6 km [5]. The construction of a complete pipeline network ensures the effective utilization of natural gas through operation scheduling.

With the competition in the natural gas market, daily natural gas management and dispatch is gathering momentum for investment. The core of ensuring the optimal dispatch of daily natural

gas is to establish an accurate dynamic model. Research on the dynamic model of natural gas pipelines has been carried out. Králik et al. [6] proposed a simplified transfer function model for natural gas pipelines, and discussed the errors introduced by the partial differential equations (PDEs) in pipeline simulation through the implicit central difference space discretization [7]. The research improved the calculation efficiency of complex natural gas pipelines. Herrán-González et al. [8] proposed two simplified dynamic models which preserved the inclination term for natural gas pipelines, and performed numerical simulations and provided a modular function library that can be re-developed. Fan et al. [9] established a natural gas pipeline model based on the mechanism equation, and utilized energy and big data analysis to evaluate the efficiency of natural gas pipelines. Behbahani-Nejad et al. [10] deduced the state space dynamic model of natural gas pipelines, and solved the problem of characterizing the complex topology of natural gas pipelines. They used a state-space model with known inlet pressure and outlet flow rate to simulate natural gas pipelines and pipeline networks and achieved good results. Alamian et al. [11] also conducted simulations of natural gas pipelines and

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pipeline networks based on state-space equations, and compared them with the formal model of the transfer function to verify the accuracy of the model. Pambour et al. [12] proposed a lumped parameter transient model for the dynamic simulation and control of natural gas pipeline networks. They introduced the developed numerical scheme and solver algorithm in detail, and benchmarked the model with other models. The research of Wen et al. [13] showed that the lumped parameter state space model was an improvement of the transfer function model by removing the control through term, making the response process of the mechanism equations closer to the actual pipeline. Fan et al. [14] adopted the control volume method for hydraulic simulation and a one-dimensional heat transfer model for transient thermal simulation. Based on this research, a complex natural gas pipeline network simulation including compressor modules and metering units can be carried out. Chaczykowski et al. [15] discussed the contributing factors to the gas temperature changes during the flow within the natural gas pipeline, and studied the effect of one-dimensional steady and unsteady state heat transfer model. In addition, some researchers have also studied the effect of key factors of natural gas pipeline networks. Behrooz et al. [16] established a dynamic model of natural gas pipeline network based on ordinary differential equations, and studied the pipeline network optimization method considering the uncertainty of customer demand. For example, Sund [17] studied the effect of ambient temperature for natural gas pipeline networks. Reddy et al. [18] used the transfer function model of the natural gas pipeline as the basis for developing a dynamic simulator of the natural gas pipeline network. The temperature variation along the pipe and unknown demands were considered for simulation and analysis.

Although there have been a lot of theoretical studies on natural gas pipeline and natural gas pipeline network simulation based on lumped models and distributed models, flow data for dynamic simulation of natural gas pipeline networks are insufficient because of few flowmeters on site. In addition, the transfer function or ordinary differential equation (ODE) models simplified based on the PDE mechanism equations are difficult to tackle and characterize the uncertainty and noise of natural gas pipelines under dynamic conditions [19]. The natural gas pipeline system is a distributed parameter system with high nonlinearity, coupling and large time delay, which is composed of multiple compressor stations and their adjustment units, so the accurate mathematical description of the pipeline transportation mechanism is very complicated [20–22]. The data-driven model approach can compensate for these problems to a certain extent [23,24]. Wei et al. [25] established the long short-term memory (LSTM) model based on singular spectrum analysis which demonstrated excellent robustness in the daily natural gas flow dynamic simulation under various characteristics of natural gas. Zhou et al. [24] conducted a novel gas turbine dynamic simulation method research based on a data-driven model of feature transfer learning via the encoding and decoding network structure, starting from a small sample and strong dynamic gas turbine data source domain. Yu et al. [26] proposed a short-term natural gas prediction combination method based on an improved back propagation neural network, and optimized the network through real-coded genetic algorithm. Su et al. [27,28] used the auto-encoder neural network to study the dynamic behavior of the natural gas pipelines system. Guarneri et al. [29] built a dynamic simulation model of the tire/suspension system when crossing road irregularities based on the recurrent neural network. Tayarani-Bathaie et al. [30] proposed a dynamic neural

network method for gas turbine dynamic modeling for real-time fault diagnosis. Hatti and Tioursi [31] constructed a dynamic neural network fuel cell control model. Lu et al. [32] proposed an optimized intelligent recurrent wavelet-based Elman neural network for dynamic simulation and optimized control of the integration of offshore wind and wave energy conversion systems.

The above literature survey results are all modeled by data-driven simulation, and good results have been achieved by using different methods on existing data. However, the data-driven approach lacks enough interpretability [33], and the neural network structure is like a black box, so we don't know how it works. Compared with the models based on the mechanism equation, it can only reflect the objective laws of the existing data [34], and it is difficult to interpret the forecasting results when encountering completely unknown data. Zhou [33] and Zhang [35] explained the convolutional neural network and formulated some learning rules. Hooker et al. [36] found that the interpretability of deep neural networks is related to sensitive areas by evaluating the quality of the input feature importance estimators.

In view of the above analysis, this paper systematically proposes an interpretable data-driven machine learning model for natural gas pipelines and pipeline networks, from the forecasting model training to dynamic simulation developing. Based on the state-space centralized model and the basic Elman neural network structure, a novel neural network model embedded with mechanism equations is proposed for dynamic simulation of natural gas pipeline networks. The major originality of this paper are summarized as follows:

- 1) The mechanism model of natural gas pipelines is extended to the form of state space equation, and the forward Euler is utilized to discretize it to obtain a fully controllable and observable natural gas pipeline model.
- 2) The natural gas pipeline model based on the form of state space equations is embedded in the machine learning model to train to obtain an interpretable Shortcut-ENN model where each neuron and weight have corresponding mechanism meanings. The interpretable machine learning model retains the principle characteristics of the mechanism equation model and reduces the uncertainty of the mechanism model and pure machine learning model.
- 3) The interpretable natural gas pipeline Shortcut-ENN model and the pipeline network model have been trained using real field measurement data from the China West-East Gas Pipeline, and have been compared with the LSTM network to explore the advantages in the mechanism interpretability and network optimization ability. The prediction results prove that the proposed model has good performance and can predict the flow rate of intermediate pipelines without measuring points.

2. Theoretical background

2.1. Natural gas pipeline mechanism model

When studying the natural gas pipeline network in this paper, it is assumed that the single natural gas pipeline meets the following requirements: (i) the natural gas flow is a constant temperature flow, which will not change the temperature of the natural gas pipeline, (ii) the friction coefficient of the pipeline and the compressibility of the pipeline section are constant. The dynamic

expression of the natural gas pipeline is obtained by applying the conservation of mass, momentum:

$$\begin{cases} \frac{c^2}{A_{cs}} \frac{\partial M}{\partial x} + \frac{\partial p}{\partial t} = 0 \\ \frac{\partial p}{\partial x} + \frac{1}{A_{cs}} \frac{\partial M}{\partial t} + \frac{pg}{c^2} \sin \theta + \frac{fc^2 M^2}{2dA_{cs}^2 p} = 0 \end{cases} \quad (1)$$

where c means the transmission speed of sound in the pipeline, M is the flow of the natural gas, A_{cs} is the cross-sectional area, x is the distance, p is the pressure, θ shows the angle between the pipeline and the horizontal plane, f is the friction coefficient and d represents the diameter of the pipeline.

The height difference of the pipeline is affected by the height of the topography, and it is not enough to affect the accuracy of the calculation of the long-distance natural gas pipeline. Here we assume that the natural gas pipelines are horizontal, that is, θ is zero degree. Utilizing the finite element method to approximate $\partial M / \partial x$ and $\partial p / \partial x$, we can get equations of the ODE form:

$$\begin{cases} \frac{dp_{out}}{dt} = -\frac{c^2}{A_{cs}} \frac{M_{out} - M_{in}}{L} \\ \frac{dM_{out}}{dt} = -\frac{A_{cs}}{L} (p_{out} - p_{in}) - \frac{fc^2 M_{in}^2}{2dA_{cs}^2 p_{in}} \end{cases} \quad (2)$$

where the subscripts *in* and *out* respectively represent the inlet and outlet of the pipeline, and L represents the length of a section of pipeline.

In natural gas pipelines, real-time gas storage is an important indicator for controlling the hydraulic and thermal balance of pipelines. The actual volume of gas in the pipeline under standard conditions is:

$$V_a = \frac{Z_0 T_0 \bar{p} A_{cs} L}{Z_a \bar{T} p_0} \quad (3)$$

where Z_a and Z_0 are the compressibility factors of natural gas under the actual working condition and standard condition, T_0 and $\bar{T}$ represent the temperature of standard condition and average temperature in the pipeline, p_0 and $\bar{p}$ represent the standard pressure and average pipeline pressure. Note that the average pipeline pressure here is assumed to be the linear average of the inlet and outlet. T_0 is generally 293.15 K here, p_0 is 101.325 KPa, and Z_0 is 1. The compressibility factor Z_a of each pipeline is obtained by taking the average pressure in the actual process and the experimental coefficient, refer to Ref. [11].

2.2. Compressor mechanism model

The centrifugal compressor is the core equipment of a natural gas booster station in the natural gas pipeline network. Compared with reciprocating compressors, centrifugal compressors are more suitable for boosting natural gas with large flow and small fluctuations. The performance of the centrifugal compressor is mainly calculated by the characteristics [37].

$$\begin{cases} M_{Cout} = \Phi_1 \left(\pi_C, \frac{n_C}{\sqrt{T_{Cin}}} \right) \\ \eta_C = \Phi_2 \left(\pi_C, \frac{n_C}{\sqrt{T_{Cin}}} \right) \end{cases} \quad (4)$$

where, the subscript C means the centrifugal compressor, the

subscripts C_{in} and C_{out} are the input and output of the compressor, π_C is the pressure ratio which is equivalent to p_{Cin}/p_{Cout} , n_C is the rotational speed of compressor, T is the temperature, η_C is the adiabatic efficiency of compressor, and $\Phi(\cdot)$ represents the mapping relationship of compressor characteristics.

Compared with the dynamics of the pipeline, the compressor has a much smaller dynamic time scale than that of the pipeline, which means that the dynamic characteristics of the compressor's flow in a short period of time can be negligible. Therefore, the natural gas pipeline network model in this paper only considers the steady-state characteristics of the compressor.

2.3. Elman neural network

Elman neural network (ENN) was proposed by Elman [35,36] and is a typical dynamic recurrent neural network. Compared with the traditional two-layer fully connected feedforward network, ENN considers the time-series information by adding a timing feedback connection in the fully connected layer. The structure of ENN is generally divided into the input layer, hidden layer, undertaking layer and output layer. Among them, the undertaking layer is used to memorize the value of hidden layer in the previous time steps, which can be considered as a single-step delay operator.

Once the input state u^t and undertake state h^{t-1} are separately linearly changed by the weighting vectors W_1 and W_2 , the hidden state h^t is obtained through the summation and the activation function $\varphi(\cdot)$, and finally the output y^t is obtained through the weight W_3 and transfer function $\varphi(\cdot)$. The mathematical expressions corresponding to ENN network are:

$$h^t = \varphi(W_1 u^t + W_2 h^{t-1}) \quad (5)$$

$$y^t = \varphi(W_3 h^t) \quad (6)$$

When the forecasting output (y_f^t) of ENN is continuous, the error function combined with the actual y_a^t at time t can be computed as:

$$e = \frac{1}{2} (y_f^t - y_a^t)^T (y_f^t - y_a^t) \quad (7)$$

After each epoch j in the training phase is processed, the weighting vectors in each layer can be updated as Eq. (8), and γ is the learning rate.

$$W_i^{j+1} \leftarrow \left(W_i^j - \gamma \frac{\partial e}{\partial W_i^j} \right), i = 1, 2, 3 \quad (8)$$

3. State space-based model for natural gas pipeline

3.1. State space model of single pipeline

The state space model is a lumped parameter model, which simplifies the system into state, input and output variables on the basis of mechanism modeling [38]. According to Eq. (2) and Eq. (3), we can construct a state space model by capturing matrix $[p_{out} \ M_{out}]^T$ as the state vector ($\mathbf{s}$) and, at the same time, define the input vector ($\mathbf{u}$) and output vector ($\mathbf{y}$) of the pipeline dynamic system as $[p_{in} \ M_{in} \ \frac{M_{in}^2}{p_{in}}]^T$ and $[p_{out} \ M_{out} \ V_a]^T$. The superscript τ here represents the transpose of the vector. Finally, Eq. (2) and Eq. (3) can be written in state space form:

$$\begin{cases} \begin{bmatrix} \dot{p}_{out} \\ \dot{M}_{out} \end{bmatrix} = \begin{bmatrix} 0 & \frac{c^2}{A_{cs}L} \\ \frac{A_{cs}}{L} & 0 \end{bmatrix} \begin{bmatrix} p_{out} \\ M_{out} \end{bmatrix} + \begin{bmatrix} 0 & \frac{c^2}{A_{cs}L} & 0 \\ \frac{A_{cs}}{L} & 0 & -\frac{fc^2}{2dA_{cs}} \end{bmatrix} \begin{bmatrix} p_{in} \\ M_{in} \\ \frac{M_{in}^2}{p_{in}} \end{bmatrix} \\ \begin{bmatrix} p_{out} \\ M_{out} \\ V_a \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ \frac{Z_0 T_0 A_{cs} L}{2Z_a \bar{T} p_0} & 0 \end{bmatrix} \begin{bmatrix} p_{out} \\ M_{out} \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ \frac{Z_0 T_0 A_{cs} L}{2Z_a \bar{T} p_0} & 0 & 0 \end{bmatrix} \begin{bmatrix} p_{in} \\ M_{in} \\ \frac{M_{in}^2}{p_{in}} \end{bmatrix} \end{cases} \quad (9)$$

$$\begin{cases} s(k+1) = (AT_s + I)s(k) + T_s Bu(k) \\ y(k) = C(AT_s + I)(AT_s + I)^{-1}s(k+1) + (D - (AT_s + I)^{-1}T_s B)u(k) \end{cases} \quad (11)$$

Therefore, $\bar{A}$, $\bar{B}$, $\bar{C}$ and $\bar{D}$ are introduced to describe the system matrix, input matrix, output matrix and direct transfer matrix of the discrete state space model according to Eq. (11).

Then the long-distance natural gas pipeline simplified mechanism equation can be described as:

$$\begin{cases} \begin{bmatrix} p_{out}^{(k+1)} \\ M_{out}^{(k+1)} \end{bmatrix} = \begin{bmatrix} 1 & -\frac{T_s c^2}{A_{cs} L} \\ -\frac{T_s A_{cs}}{L} & 1 \end{bmatrix} \begin{bmatrix} p_{out}^{(k)} \\ M_{out}^{(k)} \end{bmatrix} + \begin{bmatrix} 0 & \frac{T_s c^2}{A_{cs} L} & 0 \\ \frac{T_s A_{cs}}{L} & 0 & -\frac{T_s f c^2}{2dA_{cs}} \end{bmatrix} \begin{bmatrix} p_{in}^{(k)} \\ M_{in}^{(k)} \\ \left(\frac{M_{in}^2}{p_{in}}\right)^{(k)} \end{bmatrix} \\ \begin{bmatrix} p_{out}^{(k)} \\ M_{out}^{(k)} \\ V_a^{(k)} \end{bmatrix} = \zeta \left(\begin{bmatrix} 1 & \frac{T_s c^2}{A_{cs} L} \\ \frac{T_s A_{cs}}{L} & 1 \\ \frac{Z_0 T_0 A_{cs} L}{2Z_a \bar{T} p_0} & \frac{T_s Z_0 T_0 A_{cs}^2}{2Z_a \bar{T} p_0} \end{bmatrix} \begin{bmatrix} p_{out}^{(k+1)} \\ M_{out}^{(k+1)} \end{bmatrix} + \begin{bmatrix} -\frac{T_s^2 c^2}{L^2} & -\frac{T_s c^2}{A_{cs} L} & \frac{T_s^2 f c^4}{2dA_{cs}^2 L} \\ -\frac{T_s A_{cs}}{L} & -\frac{T_s^2 c^2}{L^2} & \frac{T_s f c^2}{2dA_{cs}} \\ \frac{Z_0 T_0 A_{cs} (L^2 - T_s^2 c^2)}{2Z_a \bar{T} p_0 L} & -\frac{Z_0 T_0 T_s c^2}{2Z_a \bar{T} p_0} & \frac{Z_0 T_0 T_s^2 f c^4}{4dA_{cs} Z_a \bar{T} p_0} \end{bmatrix} \begin{bmatrix} p_{in}^{(k)} \\ M_{in}^{(k)} \\ \left(\frac{M_{in}^2}{p_{in}}\right)^{(k)} \end{bmatrix} \right) \end{cases} \quad (12)$$

According to the basic form of the state space method, the

system matrix A can be obtained as $\begin{bmatrix} 0 & \frac{c^2}{A_{cs} L} \\ -\frac{A_{cs}}{L} & 0 \end{bmatrix}$, the input matrix B

is $\begin{bmatrix} 0 & \frac{c^2}{A_{cs} L} & 0 \\ \frac{A_{cs}}{L} & 0 & -\frac{fc^2}{2dA_{cs}} \end{bmatrix}$, the output matrix C is $\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ \frac{Z_0 T_0 A_{cs} L}{2Z_a \bar{T} p_0} & 0 \end{bmatrix}$, and the

direct transfer matrix D is $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ \frac{Z_0 T_0 A_{cs} L}{2Z_a \bar{T} p_0} & 0 & 0 \end{bmatrix}$.

According to **Rank** $[C \quad CA] = 2$ and **Rank** $[B \quad AB] = 2$, the pipeline dynamic system is judged to be completely observable and completely controllable.

3.2. Discrete state space model

Using the forward Euler method to discretize the state space equation, the discretized state space expression can be obtained:

$$\begin{cases} s(k+1) = (AT_s + I)s(k) + T_s Bu(k) \\ y(k) = Cs(k) + Du(k) \end{cases} \quad (10)$$

where T_s is the sampling period and k is the k -th discrete time point.

Substitute the $s(k)$ transformation in the state space equation into the output equation to obtain the following form:

where the coefficient ζ in the output equation is equal to item $\frac{L^2}{L^2 - T_s^2 c^2}$.

4. Interpretable Shortcut-ENN model for natural gas pipeline and network

4.1. Interpretable Shortcut-ENN model

Based on the ENN, referring to the state space form of the natural gas pipeline, a network that adds a layer of feedforward shortcut from the input layer to the output layer is defined as Shortcut-ENN, which is illustrated in Fig. 1. When the activation function is the Rectified Linear Unit (*ReLU*), the Shortcut-ENN model based on the ENN can be equivalent to this state space model of natural gas pipeline. The activation form of the *ReLU* is $\max(0, x)$, and the dataset for neural network training and testing needs to be Min-max normalization.

By embedding the mechanism equation into the Shortcut-ENN network, the complete interpretability of the machine learning model can be realized. The equivalent corresponding details are shown in Table 1. Shortcut-ENN retains the advantages of ENN network timing dynamic modeling and training, and can achieve the effect of multiple pipelines in series and multi-step time prediction. In addition, the characteristics of neural network are utilized to realize the automatic identification and dynamic update of state space parameters, which reduces the uncertainty influence.

Defining the weight of the shortcut connection layer as W_4 , according to Eq. (6), then the output of the Shortcut-ENN network can be calculated as:

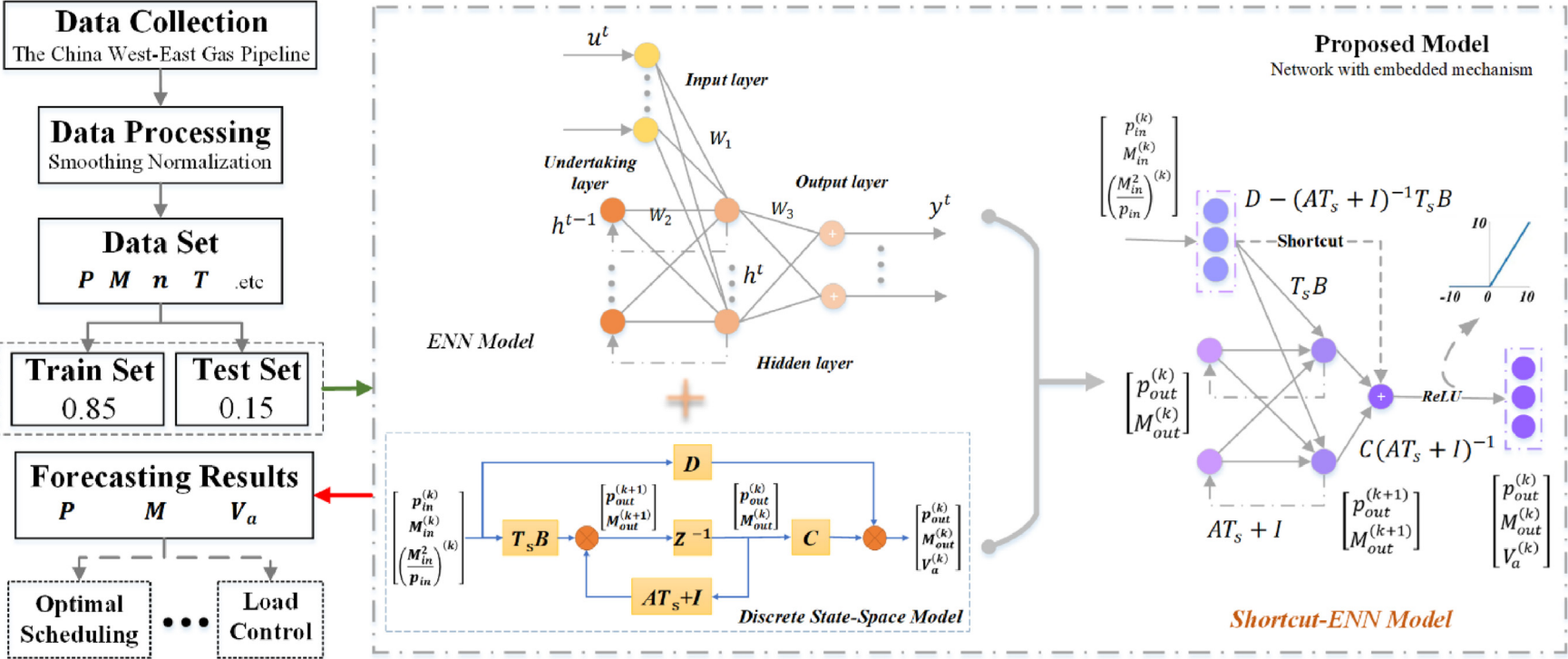


Fig. 1. The diagram of the proposed Shortcut-ENN model.

Table 1

The equivalent of the Shortcut-ENN and state space form.

Equivalence comparison	Shortcut-ENN	State space
Weights	W_1	$T_s B$
	W_2	$(AT_s + I)$
	W_3	$C(AT_s + I)^{-1}$
	W_4	$(D - (AT_s + I)^{-1}T_s B)$

$$y^t = \varphi^*(W_3 h^t + W_4 u^t) \quad (13)$$

The overall graphical process of the proposed scheme is depicted in Fig. 1. Firstly, real-time data of the China West-East Gas Pipeline Network are extracted, and data pre-processing is performed including noise reduction, abnormal point removal, and data reconciliation to obtain the dataset. The optimal forecasting results are obtained through the proposed model, and the results can be used for compressor load optimization, flow scheduling, energy consumption monitoring, etc.

In addition, since natural gas pipelines are usually long and have profile (elevation change), in a more general approach in which both temporal and spatial change in gas temperature is considered, the model interpretability of Shortcut-ENN can be weakened and the neural network can be adaptively modified when the Shortcut-ENN training effect is not good. If the distribution parameters of natural gas pipeline are sufficient, it can be considered to split a single pipeline into multiple sections which corresponds to the series connection of the Shortcut-ENN network. If not, consider adding a bias to the Shortcut-ENN model, training a compensation network, or adding a fully connected layer deepening network after the output layer to make adaptive changes to the Shortcut-ENN model.

4.2. Interpretable shortcut-ENN-based pipeline network model

In this section, a dynamic simulation of the interpretable Shortcut-ENN-based pipeline network is carried out. Usually at the actual operating site, although the flow rate is an extremely critical reference measurement data for pipeline network scheduling and operation optimization, it is difficult to ensure that all compressor stations are equipped with flow meters economically [39]. This paper establishes a dynamic pipeline network simulation model based on the constraints of the series connection of the Shortcut-ENN single pipeline model and the compressor model to simulate the flow, pressure and storage of the inlet and outlet of each pipeline.

Fig. 2 demonstrates the proposed modeling method for the natural gas pipeline network with the Shortcut-ENN, which mainly includes the pipeline module and the compressor module. Each pipeline is an independent model based on Shortcut-ENN model. The compressor module is an agent network based on meter-flow characteristic performance maps, which utilizes compressor inlet pressure, flow rate, rotation speed and outlet temperature as the network inputs, and the compressor's outlet pressure is the network output. The compressor modules were pre-trained in advance and then embedded in the pipeline network model. The compressors in the long-distance natural gas pipeline network are mainly centrifugal compressors, and their characteristic performance maps are derived from the initial measured data on site and corrected according to the operating data. The compressor characteristic performance map is mainly the relationship between speed-meter-efficiency-flow. According to the speed and meter, the

efficiency and flow can be found through the characteristic performance map.

From the perspective of data transmission in Fig. 2, the input dataset of the entire pipeline network is $p_{in,p1}, M_{in,p1}, \frac{M_{in,p1}^2}{p_{in,p1}}, (n_C, T_{Cout})_{c1,c2,\dots,c(n-1)}$, in which the subscripts $p1$ and $c1$ indicate the first pipeline and the first compressor, and the output dataset is $(M_{out}, p_{out}, V_a)_{p1,p2,\dots,pn}, (p_{out})_{c1,c2,\dots,c(n-1)}$. Note that n_C and T_{Cout} as part of the compressor input are field measurements. Except for the flow of the intermediate pipeline section due to immeasurable, the rest of the output $(p_{out}, V_a)_{p1,p2,\dots,pn}, (p_{out})_{c1,c2,\dots,c(n-1)}, (M_{out})_{pn}$ is trained as a part of the MSE loss function. As for the flow of the intermediate pipeline, it is restricted by the pressure, temperature and rotation speed of the pre-trained compressor model. Note that the input data is shown in black, the output data participating in the loss function is shown in blue, and the rest of the training data is shown in dark green.

5. Results and discussion

In this paper, a dynamic simulation model of a single pipeline and pipeline network is constructed according to the real geographical topology of the West-East Gas Pipeline, which is China's energy artery, traversing about 4200 km from east to west. And the basic parameters of related sites involved in the paper are shown in Table 2.

In order to verify the effectiveness of the proposed model, this paper utilizes LSTM-based model to compare on the same dataset that comes from the natural gas pipeline network of the West-East Gas Pipeline. Note that some missing measurement points are calculated using simulation models.

5.1. Evaluation metric

The metrics applied to both models are the mean square error (MSE), mean absolute error (MAE), regression coefficient of determination (R^2) and explained variance (EV) [40]. These indicators can be defined as follows:

$$MSE = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2 \quad (14)$$

$$MAE = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i| \quad (15)$$

$$R^2 = 1 - \frac{\sum_{i=1}^N (y_i - \hat{y}_i)^2}{\sum_{i=1}^N (y_i - \bar{y})^2} \quad (16)$$

where y_i represents the measured output, $\hat{y}_i$ is the forecasting output, $\bar{y}$ is the mean of raw data, N is the total number of the dataset, E denotes forecasting error, and E_m represents mean error.

5.2. Dynamic simulation for the single pipeline

In this section, the Shortcut-ENN and LSTM models are developed and applied to forecast the single pipeline outlet flow, pressure and pipe storage. The dataset of a pipeline from Luoning station to Zaoyang station is adopted with an interval of 1 min. The training set and the test set are divided by 0.85 to 0.15, and the total time span is 40 days. The optimal parameters of the models are

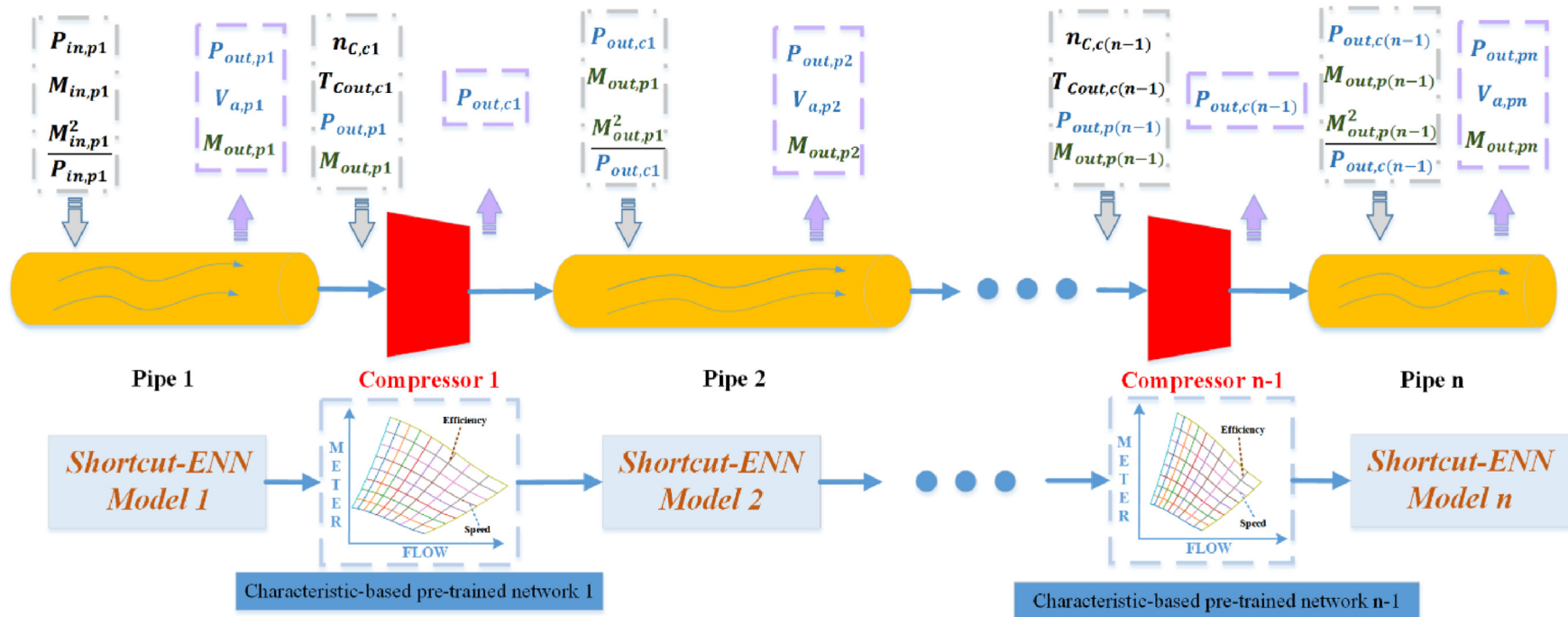


Fig. 2. The diagram of the proposed interpretable pipeline network machine learning model.

Table 2

The basic design parameters of related stations and pipelines.

Stations	Rated speed (rpm)	Rated power (KW)	Minimum inlet pressure (MPa)
Luoning	6100	31.4	4
Zaoyang	4800	29.5	6.2
Lunnan	4800	29.5	4
Kongquehe	6100	31.4	5.8
Sidaoban	4800	29.5	3.9
Shanshan	6100	31.4	5.8
Pipeline design pressure (MPa)			12

Table 3

The optimal parameters of different models.

Parameters	Shortcut-ENN	LSTM (dense)
Learning rate	0.001	0.0014
Batch size	648	512
Activation function	<i>ReLU</i>	<i>Tanh&Sigmoid</i>
Forecasting steps	30	15
Bias	False	True
Hidden layer	1	1

determined via trial and evaluation indicators of simulation and are presented in Table 3.

In order to ensure the interpretability of the Shortcut-ENN model, through multiple tests, the 30-min parameters at the pipeline inlet are trained to predict the 30-min parameters at the pipeline outlet. Since the LSTM does not require equal-dimensional inputs and outputs, through hyper-parameter tuning and error comparison, it is determined that the sliding window stride of LSTM is 15-min, and the sliding window length is 30-min. And the optimizer for both models is Adam, and the loss function for both is MSE.

The graphical comparison of the forecasting results obtained by Shortcut-ENN and LSTM model is presented in Fig. 3. Through comparison, it is found that when the layer of LSTM is the same as that of Shortcut-ENN, the forecasting results of the proposed method is significantly better than LSTM-based model. When the dense is added to LSTM, the results of prediction performance is similar to Shortcut-ENN due to the deepening of the neural network. Furthermore, it can be clearly seen from Fig. 3(a) that although LSTM can forecast the corresponding trend of the pipeline outlet pressure, the volatility of results reflected by too many training parameters is relatively large. In terms of the prediction results of trend following alone, the Shortcut-ENN model has the best effect on outlet pressure. As shown in Fig. 3(b) and (c), the following effect of the trend line at some peaks or troughs is in most cases the Shortcut-ENN model is closest to the real data.

As illustrated in Fig. 4, the statistical results of the data represented by the boxplot show that the MAE and EV of the Shortcut-ENN model are relatively concentrated, and the p_{out} and M_{out} of the MSE and EV boxplots have the smallest box line width. From the overall trend of the average line, the overall trend of MAE and EV is that p_{out} , M_{out} and V_a increase in turn. It can be observed that the average line of MAE is relatively stable, indicating that the accuracy of the Shortcut-ENN and LSTM models is similar, but the average line of EV fluctuates greatly. Comparing the three predicted parameters, the outliers of V_a are obviously the most, and the outliers of MAE's M_{out} deviate the least. As shown in Fig. 4(b), the average value of V_a is much larger than the normal value range due to the influence of large outliers, but its median value is in the middle and

lower part of the box, and the overall prediction effect can be accepted after removing outliers.

Table 4 further summarized the four evaluation criteria calculation results. According to the listed forecasting error results and Fig. 3, the R^2 of models are all higher than 0.97, and MSE are all lower than $4e-04$, which indicates that all these series have high fitting degrees to the real ones. It can be observed that all evaluation indicators of the Shortcut-ENN model for M_{out} forecasting are higher than that of the LSTM model ($MSE = 6.0572e-05$, $MAE = 0.00629$, $R^2 = 0.99536$, $EV = 0.00036$). Analysis and comparison of the EV found that the forecasting error of the proposed model has the highest interpretability, the interpretability of the p_{out} increased by about 43.5%, the M_{out} increased by about 69.7%, and the V_a increased by about 8.7%. In short, although the forecasting of the Shortcut-ENN and LSTM model for a single pipeline both have high accuracy and stability, the Shortcut-ENN model has the better variance interpretability. Note that the bold symbols in the table indicate that the method proposed in this paper has better performance than compared methods.

5.3. Dynamic simulation for the pipeline network

In this section, the natural gas pipeline network model utilized for verification of proposed model includes two compressor stations (Kongquehe station and Sidaoban station) and three pipelines of the China West-East Gas Pipeline. And Fig. 5 demonstrates the relative geographic location of the pipeline network used for dynamic simulation. The lengths of the three pipelines are approximately 220 km, 170 km, and 405 km, respectively.

The interval of pipeline network dataset is 10 min, and the data is smoothed and normalized. The training set and the test set are divided by 0.85 to 0.15, and the total time span is 40 days. Each Shortcut-ENN network imports need to be normalized by their respective true data ranges, and the exports need to be de-normalized and renormalized to fully retain the real mechanism information transmitted to the compressor model.

It can be seen from the above model training results of a single pipeline that the predictive performance of LSTM after adding a dense layer will be better than that of a single LSTM network. Therefore, in the pipeline network simulation model verification, the LSTM with a dense layer is used to compare with the Shortcut-ENN network. The overall model structure of the comparison is similar to Fig. 5, except that the Shortcut-ENN network is replaced by the LSTM network with dense. After optimization of hyper-parameters, both network models use 30 min to forecast 30 min, and the batch size is 178. The Adam optimizer learning rate of Shortcut-ENN model is 0.0013, and the learning rate of the LSTM-based model is 0.0008. The final optimization and convergence results are shown in Figs. 6, 7 and 9, including important parameters such as the p_{out} , V_a and M_{out} of each natural gas pipeline.

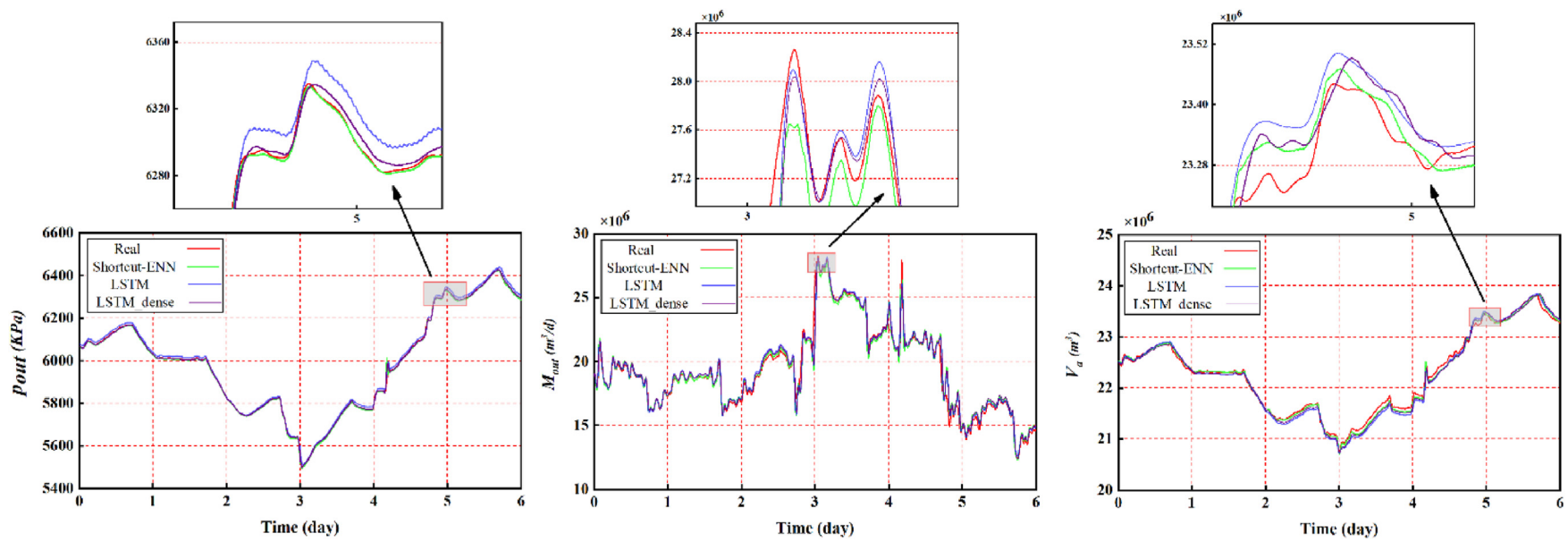


Fig. 3. Comparison of the real and predicted value of a single natural gas pipeline for 6 days.

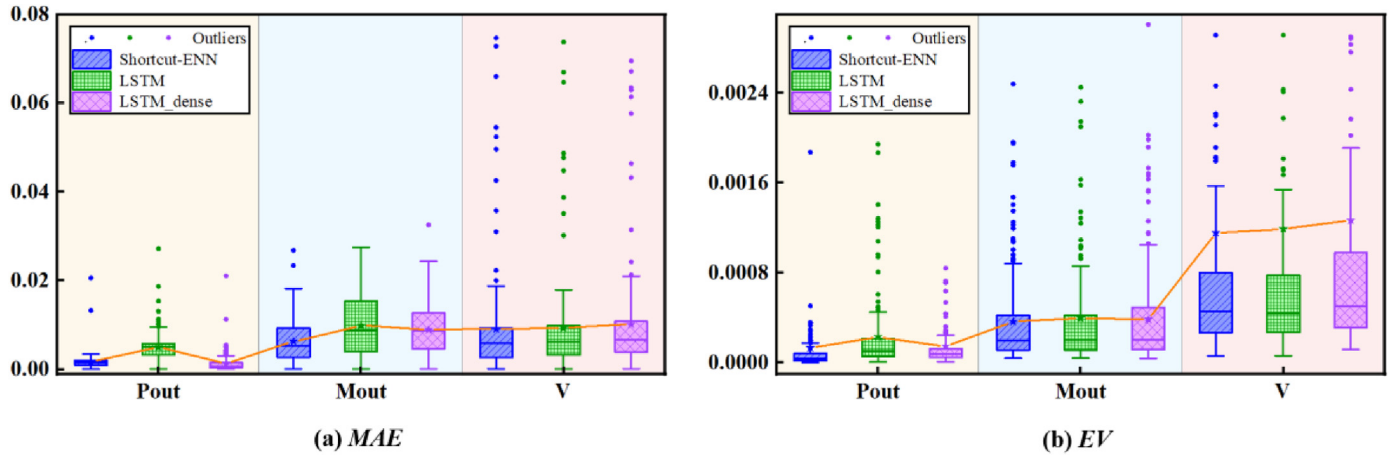


Fig. 4. Boxplots of performance metrics (MAE and EV) for different models.

Table 4

Forecasting errors of models for a single pipe model.

Indicators	Shortcut-ENN(Proposed)			LSTM			LSTM (Added dense)		
	P_{out}	M_{out}	V_a	P_{out}	M_{out}	V_a	P_{out}	M_{out}	V_a
MSE	8.4737e-06	6.0572e-05	3.1022e-04	3.8348e-05	1.4574e-04	3.1017e-04	6.6084e-06	1.0934e-04	3.4342e-4
MAE	0.00169	0.00629	0.00910	0.00501	0.00988	0.00933	0.00126	0.00894	0.01017
R^2	0.99944	0.99536	0.97776	0.99747	0.98884	0.97766	0.99956	0.99163	0.97539
EV	0.00013	0.00036	0.00115	0.00023	0.00040	0.00119	0.00015	0.00038	0.00126

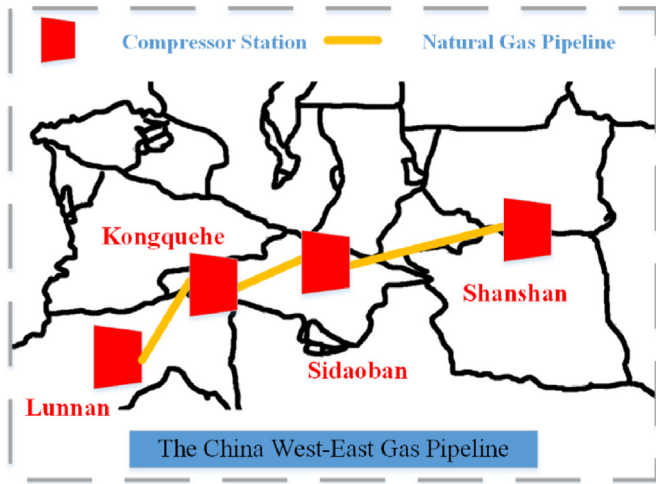


Fig. 5. Physical layout between compressor station and compressor units of dynamic simulation pipeline network.

From the predicted trend shown in Fig. 6, the red line represents the real data, the blue line represents the Shortcut-ENN-based model prediction result, and the green line represents the LSTM-based model prediction result. Compared with LSTM-based models, forecasted values of the Shortcut-ENN model have the higher degree of accuracy to real values of the $p_{out, p1, p2, p3}$, $V_a, p1, p2, p3$ and $M_{out, p3}$. The accuracy depends on both quantity and direction. In other words, there are both a smaller quantitative difference and a more consistent trend between the real value and

the forecasted value at each time step. In addition, it can be found from Fig. 6 that with the connection of the Shortcut-ENN network, there is no obvious error in the prediction results of the subsequent pipeline, but the error of the downstream pipeline of the LSTM-based model is increasing. To compare the prediction accuracy further quantitatively under different models, Table 5 shows the prediction error results under different indicators. Unlike the prediction results of the single pipeline model, it can be observed that all the four evaluation criteria (MSE, MAE, R^2 , EV) of Shortcut-ENN are better than LSTM with dense model. Among the different prediction parameters, the prediction accuracy of outlet pressure is the highest, and the prediction accuracy of outlet flow rate is the lowest. In addition, the overall prediction effect of the previous pipeline will be better than that of the subsequent pipeline. From the perspective of model inputs, since the pressure and flow information are the only most accurate given by the first pipeline, the errors of the subsequent pipelines will continue to accumulate due to error propagation. Comparing the average prediction errors, MSE, MAE, R^2 and EV increased by approximately 84.4%, 60.1%, 0.75%, and 53.3%, respectively.

Fig. 7 shows the comparison between the real flow rate $M_{out, p3}$ at the outlet of the last pipeline of the pipeline network and the flow trend predicted by the different models. By comparing the Shortcut-ENN-based model and the LSTM-based model, it can be found that from the first day to the third day, the proposed model trend's prediction effect is better than that of the LSTM-based model. On the fourth day, the Shortcut-ENN-based model deviates from the LSTM-based model, but the trend soon stabilized. On the last day, the prediction results of the LSTM-based model deviates significantly from the real value, while the Shortcut-ENN-based model has good prediction performance. From the analysis in

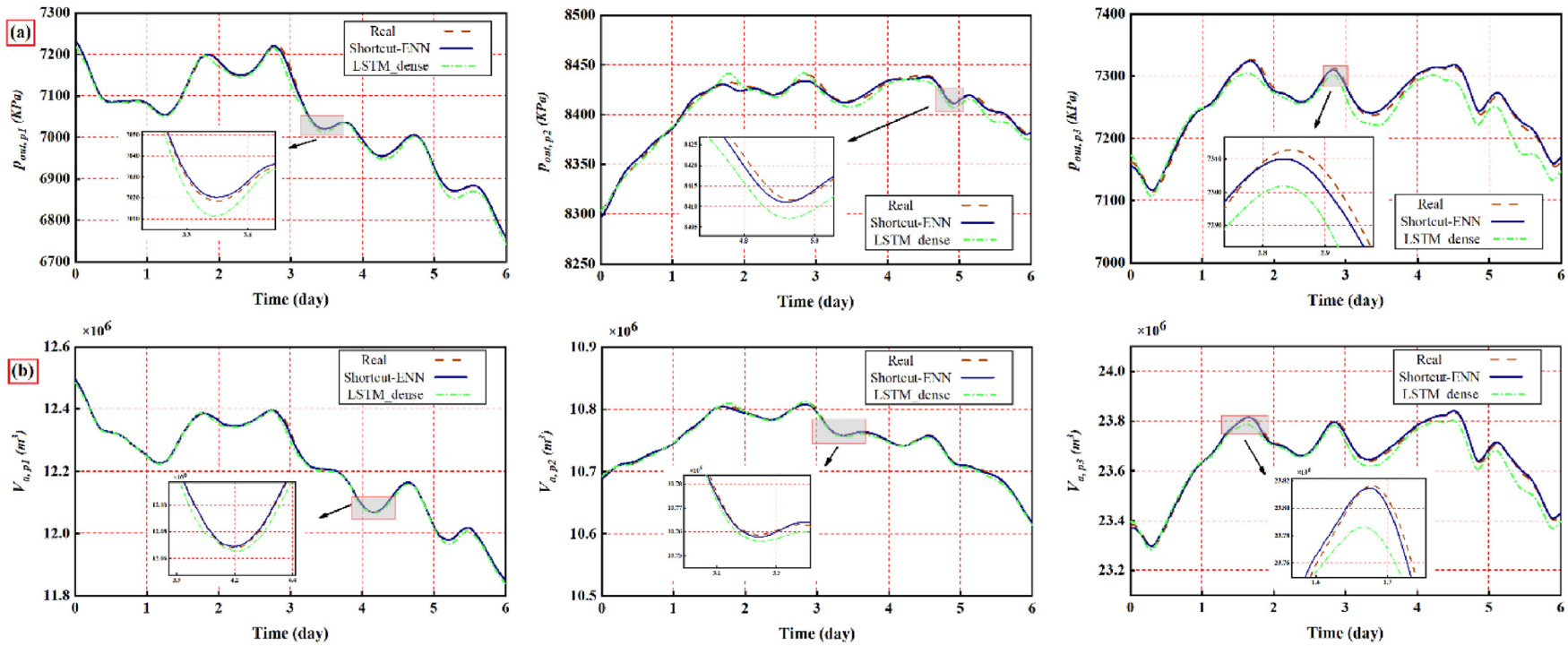


Fig. 6. Comparison of the actual and predicted value of the natural gas pipeline network.

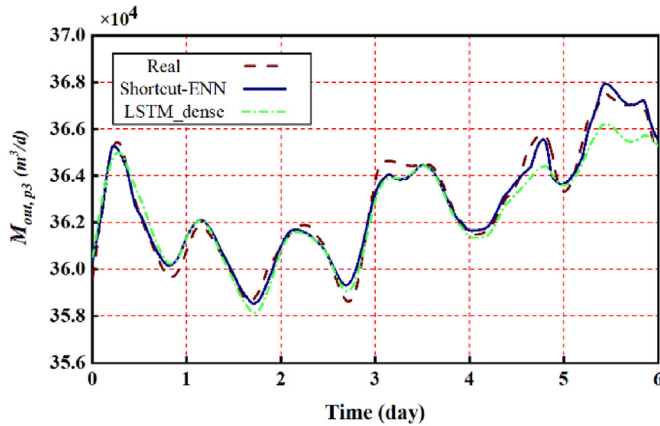


Fig. 7. Comparison of the $M_{out, p3}$ of different models.

Table 5, the MSE obviously has the largest improvement span, and the different models' daily forecast errors of this evaluation indicator are presented in Fig. 8. The prediction results of the Shortcut-ENN model are relatively stable every day, and the prediction generalization of the LSTM-based model is worse than that of the proposed model. It can be seen from Fig. 8 that the Shortcut-ENN model's prediction accuracy of $p_{out, p2}$ and $V_{a, p2}$ on the second day and $M_{out, p3}$ on the third day is not as good as that of the LSTM-based model, but the gap is small. And the Shortcut-ENN-based model's prediction effect of all parameters of all days of the first pipeline is better than that of LSTM-based model. Therefore, by comparing the trend, accuracy, and generalization of different models, it can be considered that the Shortcut-ENN model has a better predictive effect on the natural gas pipeline network, and the intermediate pipeline can be considered to represent the real flow rate by the mechanism constrained predicting flow rate.

As shown in Fig. 9, the blue curves represent the prediction results of the unmeasurable flow rate at the outlet of the first pipeline and the second pipeline, and the green square represents the time delay under several typical dynamic conditions. The overall trend of the flow rate between different pipelines is similar. The strong dynamic change of the compressor and the long-distance transportation cause these uncertain time delay. The parameter sharing, embedded pipes and compressor mechanism during entire natural gas pipeline network training process aim to eliminate the uncertainty of data and the influence of data without measurement points to a certain extent, especially the flow rate. Note that if the amount of training data is large enough, the

Shortcut-ENN model can be directly used for parameter sharing training, which can eliminate the uncertainty caused by the time delay to a certain extent. If not, the Shortcut-ENN network input layer needs to add a fully connected layer to deepen the network to compensate for uncertainty. Thus, it is fair to say that the Shortcut-ENN model exhibits a relatively better performance in pipeline network forecasting and simulation.

6. Conclusions

Aiming at natural gas pipeline networks with lack of measuring points and uncertain large time delay, this paper starts from the principle and derives through the natural gas pipeline mechanism formula. An interpretable natural gas pipeline and pipeline network machine learning model is proposed based on the Shortcut-ENN whose intermediate nodes have clear meanings, which can support the joint representation of mechanism knowledge and data features in the model. Embedding the mechanism characteristics into the machine learning model not only improves the interpretability of the model, but also eliminates the uncertainty of multi-parameters to a certain extent. And this model is designed to accurately simulate natural gas pipelines and pipeline networks and solve the problem of no flow rate metering stations in the intermediate pipelines. In order to verify the interpretability and dynamic simulation performance of the proposed model, a single pipeline and a pipeline network were compared and verified using real datasets of the China West-East Gas Pipeline. Finally, through the analysis of the trend, accuracy and generalization of the forecasting results, an interpretable Shortcut-ENN machine learning model suitable for dynamic pipeline and pipeline network simulation is proposed. Based on the research and analysis in this paper, the key findings are as followings:

- (1) By involving the natural gas pipeline mechanism information and compressor mechanism constraints, the natural gas pipeline network dynamic simulation model based on Shortcut-ENN can effective predict the pressure, volume and flow rate of the intermediate pipeline without the flow rate metering station. Comparing the average prediction errors with LSTM-based model, MSE, MAE, R^2 , EV of proposed model increased by approximately 84.4%, 60.1%, 0.75%, and 53.3%, respectively, and the R^2 is about larger than 0.99, and the EV is about less than 0.02.
- (2) The intermediate hidden elements and weights of the neural network model based on Shortcut-ENN have interpretable physical meanings and expressions. Under the uncertainty of time delay, the interpretable natural gas pipeline network

Table 5

Forecasting errors of the Shortcut-ENN (Proposed) and LSTM (Added dense) model for pipeline network simulation.

Models	Indicators	$p_{out, p1}$	$V_{a, p1}$	$p_{out, p2}$	$V_{a, p2}$	$p_{out, p3}$	$V_{a, p3}$	$M_{out, p3}$	Average
Shortcut-ENN (Proposed)	MSE	4.4409e-06	1.3199e-05	7.5105e-06	6.7500e-05	7.7781e-06	0.00015	0.00044	0.00010
	MAE	0.00145	0.00250	0.00221	0.00663	0.00235	0.01027	0.01714	0.00608
	R^2	0.99967	0.99989	0.99658	0.99934	0.99746	0.99916	0.98915	0.99732
	EV	0.00201	0.00346	0.00264	0.00805	0.00245	0.01073	0.02106	0.00720
LSTM	MSE	5.14702e-05	0.000153	2.1508e-05	0.00019	0.00013	0.00246	0.00143	0.00063
	MAE	0.00484	0.00835	0.00383	0.01143	0.00908	0.03976	0.02946	0.0153
	R^2	0.99759	0.99921	0.99334	0.99911	0.97067	0.99060	0.97846	0.98985
	EV	0.00558	0.00961	0.00371	0.01109	0.00821	0.03593	0.03387	0.015423

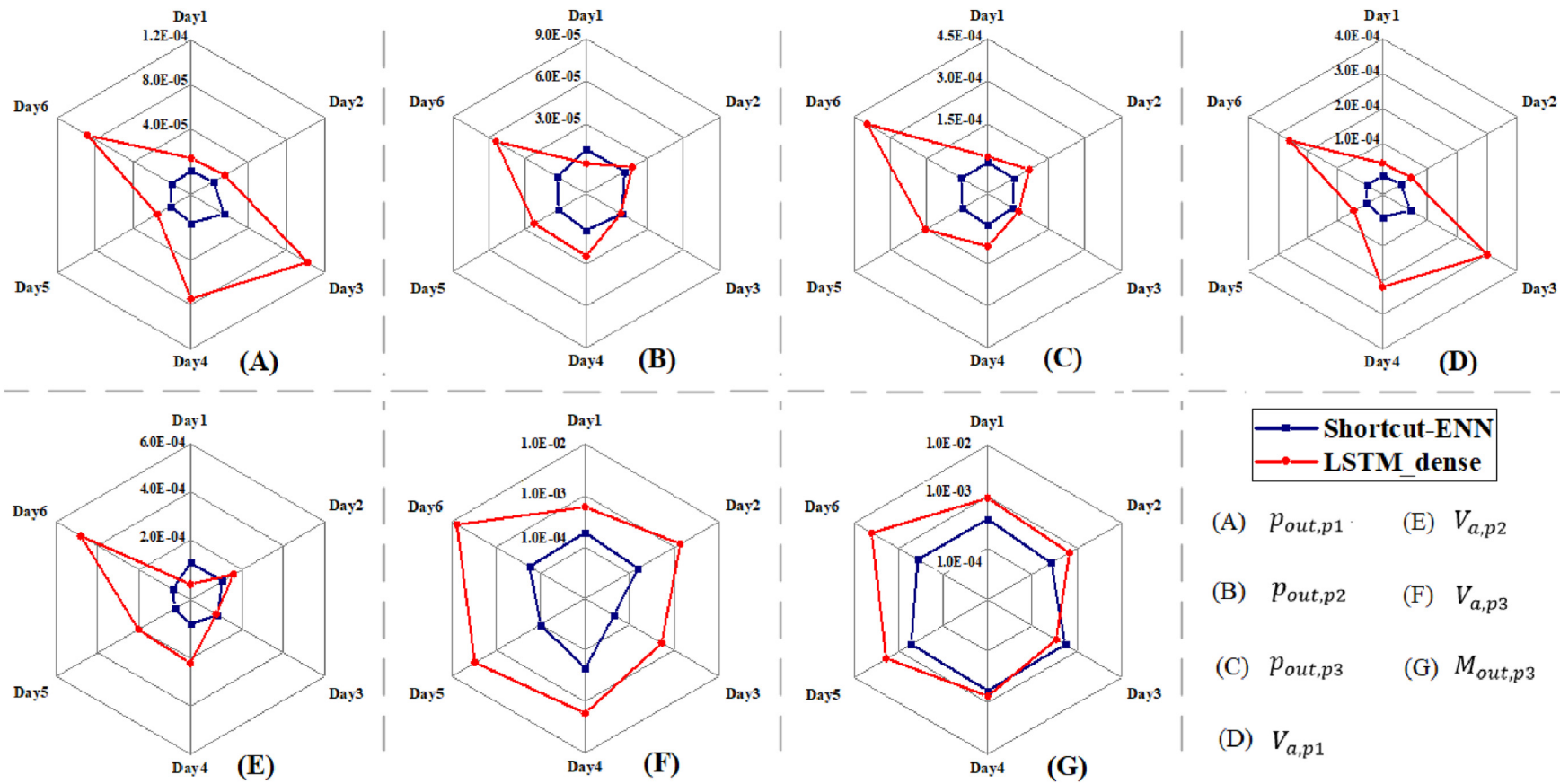


Fig. 8. MSE performance of different models for natural gas pipeline network simulation.

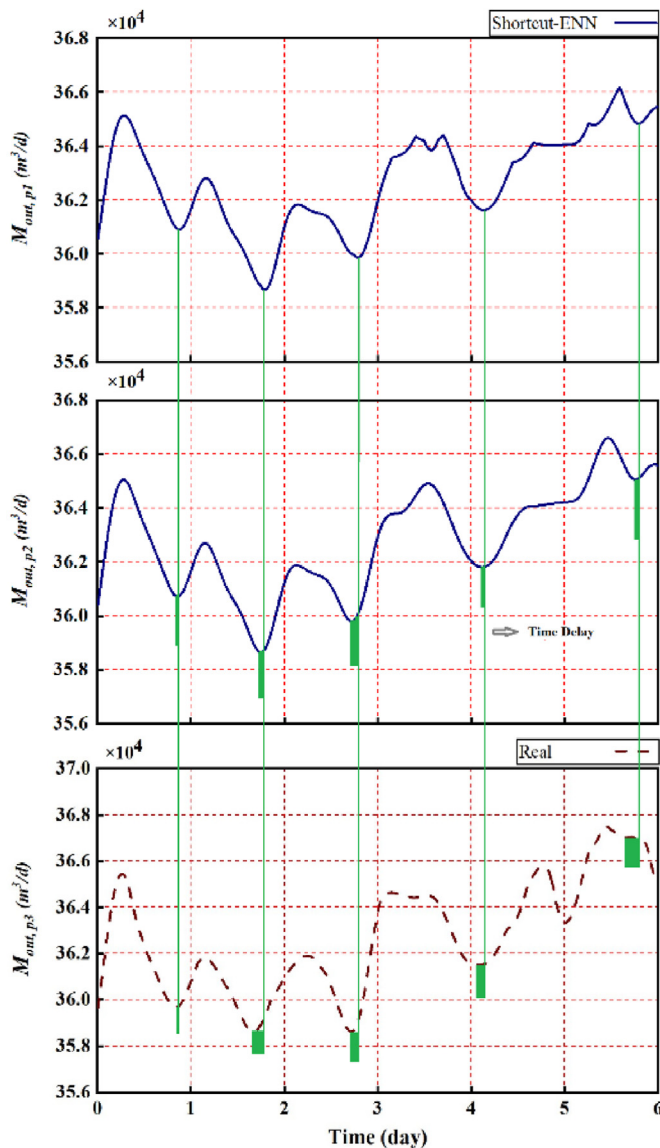


Fig. 9. Predicted $M_{out, p1}$, $M_{out, p2}$ by Shortcut-ENN model and real $M_{out, p3}$

machine learning model obtained by training a large amount of data can effectively carry out dynamic simulation, but in the absence of sufficient data, it may be necessary to add a fully connected layer behind the Shortcut-ENN network to compensate for uncertainty. Furthermore, each pipeline model is an independent modular unit based on Shortcut-ENN, which can be well embedded and integrated into larger pipeline network systems.

- (3) For further research, the definition of the weight of loss function and the scope of parameter sharing of Shortcut-ENN model need to be analyzed and researched in a more complex pipeline network, and the temperature and compressibility changes of the pipeline in temporal and spatial can be considered for embedding in the machine learning model. In addition, in order to further verify its generalization on a larger scale, this conclusion need to be proved by more case studies.

Credit authorship contribution statement

Dengji Zhou: Conceptualization, Modeling, Methodology, Formal analysis, Writing – original draft. Xingyun Jia: Data curation, Methodology, Investigation, Writing – original draft. Shixi Ma: Modeling, Writing – review & editing. Tiemin Shao: Writing – review & editing, Methodology. Dawen Huang: Resources, Writing – review & editing. Jiarui Hao: Investigation, Writing – review & editing. Taotao Li: Reviewing, Editing and Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Nomenclature

PDE: partial differential equation
ODE: ordinary differential equation
LSTM: long short-term memory
ENN: Elman neural network
Shortcut-ENN: Elman neural network with forward shortcut
M: the flow of natural gas
 $\Phi(\cdot)$: mapping relationship of compressor
 π_c : pressure ratio
 n_c : rotational speed of compressor
T: temperature
 η_c : adiabatic efficiency of compressor
c: transmission speed of sound
 A_{cs} : cross-sectional area
 u^i, h^i, y^i : input, hidden and output state
W: weighting vector
 $\varnothing(\cdot), \varphi(\cdot)$: activation and transfer function
 y_f^i, y_a^i : forecasting and actual output of ENN
e: error function
j: epoch
 γ : learning rate
C: compressor
L: length of the natural gas pipe
n: quantity
x: distance
p: pressure
 θ : angle between pipeline and horizontal plane
f: friction coefficient
d: diameter of pipeline
 Z_a, Z_0 : compression factors
 T_0, p_0 : standard condition temperature and pressure
 $\bar{T}, \bar{p}$: average temperature and pressure
A, B, C, D: system, input, output and forward matrix
 T_s : sampling period
k: the *k*-th discrete time point
 $\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D}$: matrices of discrete state space model
 ζ : coefficient of output equation
MSE: mean square error
MAE: mean absolute error
 R^2 : regression coefficient of determination
EV: explained variance
 $y_i, \hat{y}_i$: measured and forecasting output
 $\bar{y}_i$: mean of raw data
N: number of the dataset
 E, E_m : forecasting and mean error